

Pure Pursuit vs. Stanley for DV Autocross

Lateral controller comparison in MATLAB / Simulink · Formula Student Driverless · Alex Demir

WHAT I DID

I built one Simulink model of a driverless Formula Student car driving a lap of a cone-marked Autocross track, and ran that same model twice: once steering with pure pursuit, once with Stanley. Everything else was kept identical, including the track, the speed the car was told to drive, the throttle and brake controller, the steering actuator with its speed and angle limits, the 100 Hz control rate and the solver. Only the steering law changed between the two runs, so any difference in the results comes from the steering law and nothing else.

The track is a closed loop marked by blue cones on the left and yellow on the right. It is 212 m long and 3.5 m wide, with a long straight into a tight hairpin, a slalom, a corner that tightens as you drive through it, and one fast open bend. I picked those features on purpose, because they are where the two controllers behave differently. The layout follows the size limits the Formula Student rules give for this event.

Each controller has two settings, tuned the same way for both: a grid search scored by the same measure, tested beyond the normal operating range so that settings which only look good in easy conditions do not win. I also wrote 21 tests that check each steering law against answers that can be worked out by hand, and one of them found a real mistake in my pure pursuit code. I checked the Simulink model against a second, independent implementation of the same maths as well, and the two agree to eleven decimal places.

WHAT I MEASURED, AND WHAT CAME OUT

Measure	Pure Pursuit	Stanley
Lap time	17.96 s	18.18 s
Worst distance off the line	0.90 m	0.13 m
Average distance off the line	0.41 m	0.06 m
How hard the steering works	8.7 °/s	21.1 °/s
Cones knocked over	0	0
Competition points	100.0	98.9

Pure pursuit was 0.22 s quicker and scored 1.1 points more, and its steering worked less than half as hard. Stanley followed the line about seven times more closely. Neither hit a cone.

The lap time surprised me at first, because pure pursuit follows the line far less accurately. The reason turned out to be simple. Pure pursuit aims at a point some distance ahead of the car, so in every corner it cuts across the inside of the line. Over a lap it drove 208.5 m where Stanley drove 210.2 m. Both cars were told to drive at the same speed, so the one covering less ground finishes sooner. When I forced pure pursuit to follow the line as tightly as Stanley, it became the slower of the two. Its advantage was the shortcut, not better control.

That also means the distance-off-the-line numbers should not be read as pure pursuit being bad. The line the car follows is the middle of the track, and the middle is not the fast way round. A racing driver cuts corners on purpose. What those numbers really tell you is how much room is left before the car reaches the cones.

WHAT HAPPENS ON A NARROWER TRACK

All of the above was on a 3.5 m track. The rules allow the organisers to make it as narrow as 3 m, so I repeated the runs at 3.25 m and 3.00 m. Knocking a cone costs 2 seconds.

Track width	Pure pursuit: cones / points	Stanley: cones / points
3.50 m	0 / 100.0	0 / 98.9
3.25 m	0 / 100.0	0 / 98.9
3.00 m	2 / 80.3	0 / 100.0
3.00 m, 30% faster	6 / 50.5	0 / 100.0

The reason is how much room each one leaves. On the 3.5 m track pure pursuit passed the cones with 25 cm to spare, and only 11 cm when driven 30% faster. Stanley had over a metre. Take half a metre off the track and pure pursuit's margin is gone: it starts hitting cones and loses 20 to 50 points, while Stanley still had 77 cm of room and never touched one at any width or speed I tried. I also tested both with a delayed and a noisy position estimate, and once tuned properly they handled that equally well.

CONCLUSION

If I had to put one of these two on the car for an Autocross run, I would choose Stanley. Not because it is faster, it is not, but because it keeps its distance from the cones. Pure pursuit buys its 0.22 s by driving close to the track edge, and that only pays off as long as nothing else goes wrong. On a real car something always does: the position estimate drifts, a cone is detected in the wrong place, the surface is damp. Stanley has 77 cm of room for that. Pure pursuit has 25, and none at all on a narrow track.

The honest answer, though, is that neither of them really wins. Both are purely geometric. They look at where the car sits relative to a line and turn the wheel accordingly, and neither knows anything about the car itself: no mass, no tyres, no grip, no notion that the rear can step out. That is acceptable at Autocross speeds on a wide track, and it is why this comparison came down to small margins rather than one controller being right. Push the same two laws to the limit of grip, or onto a genuinely fast car, and both fail the same way. They keep asking for a steering angle the tyres cannot deliver. What is needed there is a controller built on a model of the vehicle, such as LQR or MPC on a dynamic bicycle model with tyre stiffness, which can anticipate the car sliding instead of only reacting once it has. Of the two I tested, pure pursuit is the one I would trust least as speed rises, because it looks further ahead the faster it goes, so it cuts corners hardest where the margin matters most.